kth constraint will act in the transverse direction to that constraint, i.e. along the direction of the gradient ∇g_k, and the equations of motion may be written

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{4.4}$$

$$\dot{\boldsymbol{p}} = -\nabla U(\boldsymbol{q}) - \sum_{k=1}^{l} \lambda_k \nabla g_k(\boldsymbol{q}), \tag{4.5}$$

$$0 = g_j(\boldsymbol{q}), \qquad j = 1, 2, \ldots, l, \tag{4.6}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_l$ are the Lagrange multipliers. It is often convenient to combine all the constraints and to describe them by a single vector-valued function $\boldsymbol{g} : \mathbb{R}^{\mathrm{N}_c} \to \mathbb{R}^l$, then we may write the constrained equations of motion in the form

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{4.7}$$

$$\dot{\boldsymbol{p}} = -\nabla U(\boldsymbol{q}) - \boldsymbol{g}'(\boldsymbol{q})^T \boldsymbol{\lambda}, \tag{4.8}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{q}). \tag{4.9}$$

Here $\boldsymbol{\lambda}$ is an l-dimensional vector of Lagrange multipliers.

Constraints such as those described above define a manifold in the configuration space $\mathcal{M} = \{\boldsymbol{q} | \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{0}\}$. Given a motion $(\boldsymbol{q}(t), \boldsymbol{p}(t), \boldsymbol{\lambda}(t))$ of (4.7)–(4.9) we note that $\boldsymbol{g}(\boldsymbol{q}(t)) \equiv \boldsymbol{0}$ at all times t, hence, by differentiation

$$\boldsymbol{g}'(\boldsymbol{q}(t))\dot{\boldsymbol{q}}(t) \equiv \boldsymbol{0},$$

which we may rephrase as

$$\boldsymbol{g}'(\boldsymbol{q}(t))\boldsymbol{M}^{-1}\boldsymbol{p}(t) \equiv \boldsymbol{0}.$$

Thus while $\boldsymbol{q} \in \mathcal{M}$, $\boldsymbol{q}$ and $\boldsymbol{p}$ together satisfy a separate constraint relationship $\boldsymbol{g}'(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p} = \boldsymbol{0}$. This is sometimes referred to as a *co-tangency condition* or the *hidden constraint*. The set of pairs $(\boldsymbol{q}, \boldsymbol{p})$ satisfying, simultaneously, the equations

$$\boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{0}, \qquad \boldsymbol{g}'(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p} = \boldsymbol{0},$$

is called the *co-tangent bundle* associated to the manifold $\mathcal{M}$, and we refer to it as $T^*\mathcal{M}$. The co-tangent bundle is the collection of valid positions and momenta for a holonomically constrained mechanical system; it is the constrained analog of the phase space. The word "co-tangent" refers to the fact that velocities $\boldsymbol{v} = \dot{\boldsymbol{q}}$ lie in the tangent space $\boldsymbol{g}'(\boldsymbol{q})\boldsymbol{v} = \boldsymbol{0}$, whereas the momenta are viewed as "co-vectors," that is one-forms or linear mappings taking tangent vectors to scalars (through the Euclidean inner product).

The solution of (4.7)–(4.9) evolves on the co-tangent bundle. Along a solution we have $\boldsymbol{g}'(\boldsymbol{q}(t))\boldsymbol{M}^{-1}\boldsymbol{p}(t) \equiv \boldsymbol{0}$. If we differentiate this again, we obtain

$$\boldsymbol{g}'(\boldsymbol{q}(t))\boldsymbol{M}^{-1}\dot{\boldsymbol{p}}(t) + \boldsymbol{\varphi} \equiv \boldsymbol{0}, \tag{4.10}$$